

Retail Customer Analysis

Phase 6 — Market Basket Analysis

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Overview

This phase identifies which products customers buy together, using association-rule mining (the Apriori algorithm, cross-checked with FP-Growth). The output is a set of rules of the form "baskets containing A also tend to contain B", ranked by lift — how much more often two products are bought together than chance predicts. To keep the analysis fast and statistically reliable, it focused on the 120 most-purchased products at a moderate frequency threshold; rare products were excluded as a pair seen a handful of times is not a dependable pattern.

Results at a Glance

15x TOP ASSOCIATION STRENGTH (LIFT)	242 ASSOCIATIONS DISCOVERED	42 STRONG RULES (CONF. ≥ 50%)	25,814 MULTI-ITEM BASKETS	120 TOP PRODUCTS ANALYSED
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Key Findings

1. Matching product ranges are the strongest driver of co-purchases.

The Regency tea range — the 3-tier cakestand with its rose and green teacups and saucers — tops every measure: these pieces are bought together up to 15 times more often than chance. Customers buy the set, not the single piece.

2. Colour and theme variants sell as pairs.

Customers who buy one colour of the Bakelike alarm clock frequently buy the other (green and red, 13x); the same holds for themed pairs such as the Spaceboy and Dolly Girl lunch boxes and the owl and Scotty-dog hand warmers.

3. These associations are directly actionable.

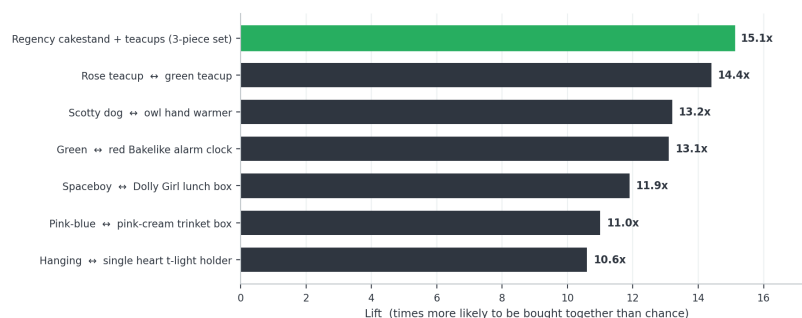
The strongest rules point to concrete moves: bundle the Regency tea pieces, recommend the matching colour at checkout, and place themed variants side by side. 42 rules are strong enough (50%+ confidence) to act on with confidence.

4. The method was chosen for reliability, not just coverage.

Focusing on the top 120 products at a moderate support floor produced a manageable 242 associations rather than thousands of noisy ones, and two independent algorithms (Apriori and FP-Growth) returned identical results — confirming the patterns are real, not artifacts.

Strongest Product Associations

Top co-purchase associations by lift — a lift of 15 means the pair is bought together 15x more often than chance.



What's Next — Phase 7

Phase 7 brings the project together into an interactive Tableau dashboard, drawing on every output now in the database — sales, cohorts, segments, CLV, and these basket rules — for a single decision-ready view.